

EE-312L Signals and Systems Lab
Laboratory Session 04
**Understanding The Continuous-Time Signals
and Their Transformation Using MATLAB**

October 16, 2024

Outlines:

1. Objective
2. Unit Step Function
3. Rect Function
4. Unit Ramp Function
5. Time Shifting
6. Time Scaling
7. Time Reversal
8. Amplitude Scaling

1 Objective

The objective of this lab session is to understand continuous-time signals and their transformation using MATLAB.

2 Continuous-Time Signals

2.1 Signum Function

$$\text{sgn}(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

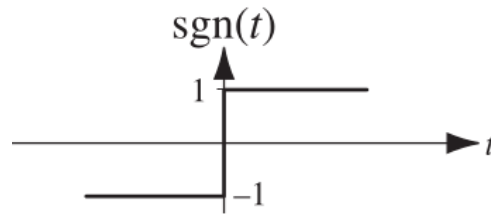


Figure 1: Signum function

The `sign(t)` function in MATLAB can be used to generate the plot of signum function.

```
1 t = -5:0.01:5;  
2 y = sign(t);  
3 plot(x,y, 'LineWidth',2.5)  
4 xlabel('t');  
5 ylabel('sgn(t)');  
6 title('Signum function plot using MATLAB');  
7 ylim([-2 2])  
8 xlim([-5 5])  
9 grid minor on
```

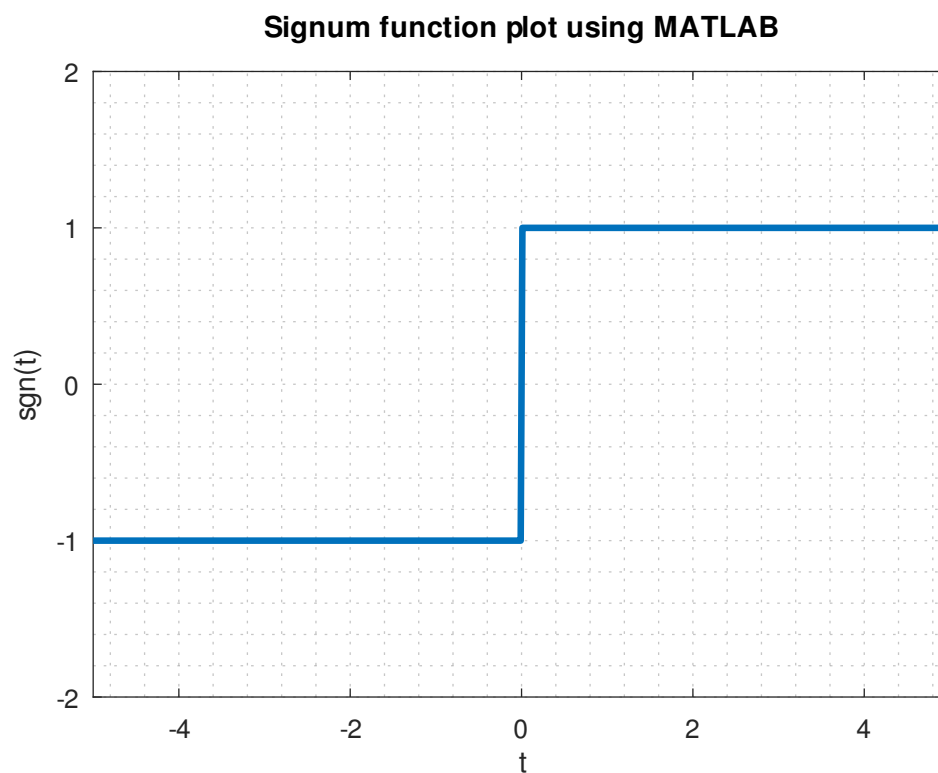


Figure 2: Signum function using MATLAB

2.2 Unit-Step Function

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

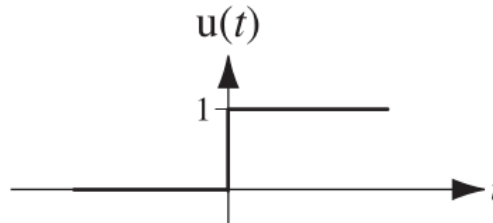


Figure 3: Unit Step function

2.3 Unit-Ramp Function

$$u(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

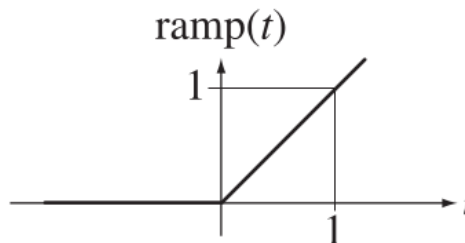


Figure 4: Unit Ramp function

2.4 Rectangular Function

$$rect(t) = \begin{cases} 1, & |t| \leq \frac{1}{2} \\ 0, & |t| > \frac{1}{2} \end{cases}$$

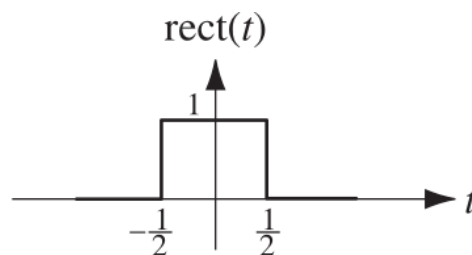


Figure 5: Rect function

2.5 Triangular Function

$$tri(t) = \begin{cases} 1 - 2|t|, & |t| \leq \frac{1}{2} \\ 0, & \text{Otherwise} \end{cases}$$

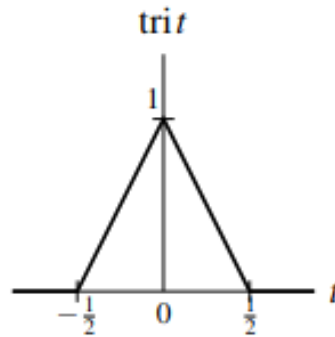


Figure 6: Tri function

3 Generating Continuous-Time Signals Using MATLAB

We will be using the following functions:

3.1 Function for generating unit step function

```
1 function y = ustep(t, ad)
2     % t: time
3     % m: slope
4     % ad: advance (for +ve value) delay(for -ve value),
5     N = length(t);
6     y = zeros(1,N);
7     for i = 1:N,
8         if t(i) >= -ad,
9             y(i) = 1;
10        endif
11    endfor
```

3.2 Function for generating unit ramp function

```
1 function y = ramp(t, m, ad)
2     % t: time
```

```
3 % m: slope
4 % ad: advance (for +ve value) delay(for -ve value),
5 N = length(t);
6 y = zeros(1,N);
7 for i = 1:N,
8     if t(i) >= -ad,
9         y(i) = m*(t(i) + ad);
10    endif
11 endfor
```

3.3 Function for generating rect function

```
1 function y = rect(t, td)
2 % t: time
3 % m: slope
4 % ad: advance (for +ve value) delay(for -ve value),
5 N = length(t);
6 y = ustep(t,0.5+td) - ustep(t,-0.5+td);
```

3.4 MATLAB code for generating plots

```
1
2 clear all, clc, close all;
3 t = -5:0.01:5;
4
5 figure(1)
6 plot(t,ustep(t,0),'r', 'linewidth',2.5);
7 title('Unit step signal u(t)')
8 axis([-5 5 -3 3]);
9 legend('Unit Step');
10 xlabel('t');
11 ylabel('u(t)');
12 grid minor on;
13 saveas(gcf,'plots/plot01.pdf')
14
15 figure(2)
16
17 subplot(2,1,1)
18 plot(t,ustep(t,0),'r', 'linewidth',2.5);
19 axis([-5 5 -3 5]);
20 legend('Unit Step');
21 xlabel('t');
22 ylabel('u(t)');
```

```
23 grid minor on;
24
25 subplot(2,1,2)
26 plot(t,ustep(t,2),'r', 'linewidth',2.5);
27 axis([-5 5 -3 5]);
28 legend('Unit Step advanced by 2 units');
29 xlabel('t');
30 ylabel('u(t+2)');
31 grid minor on;
32 saveas(gcf,'plots/plot02.pdf')
33
34 figure(3)
35
36 subplot(2,1,1)
37 plot(t,ustep(t,0),'r', 'linewidth',2.5);
38 axis([-5 5 -3 5]);
39 legend('Unit Step');
40 xlabel('t');
41 ylabel('u(t)');
42 grid minor on;
43
44 subplot(2,1,2)
45 plot(t,ustep(t,-1),'r', 'linewidth',2.5);
46 axis([-5 5 -3 5]);
47 legend('Unit Step delayed by 1 unit');
48 xlabel('t');
49 ylabel('u(t-1)');
50 grid minor on;
51 saveas(gcf,'plots/plot03.pdf')
52
53
54
55 %%% Unit React Function
56
57
58 figure(4)
59
60 subplot(3,1,1)
61 plot(t,ustep(t,0.5),'r', 'linewidth',2.5);
62 axis([-5 5 -3 3]);
63 legend('Unit Step advanced by 0.5 units');
64 xlabel('t');
65 ylabel('u(t+0.5)');
66 grid minor on;
67
```

```
68 subplot(3,1,2)
69 plot(t,ustep(t,-0.5),'r','linewidth',2.5);
70 axis([-5 5 -3 3]);
71 legend('Unit Step delayed by 0.5 units');
72 xlabel('t');
73 ylabel('u(t-0.5)');
74 grid minor on;
75
76 subplot(3,1,3)
77 plot(t,ustep(t,0.5)-ustep(t,-0.5),'r','linewidth',2.5);
78 axis([-5 5 -3 3]);
79 legend('rect function');
80 xlabel('t');
81 ylabel('rect(t)');
82 grid minor on;
83 saveas(gcf,'plots/plot04.pdf')
84
85
86 figure(5)
87
88 plot(t,rect(t,0),'r','linewidth',2.5);
89 axis([-5 5 -3 3]);
90 legend('unit rect function');
91 xlabel('t');
92 ylabel('rect(t)');
93 grid minor on;
94 saveas(gcf,'plots/plot05.pdf')
95
96
97 figure(6)
98 subplot(2,1,1)
99 plot(t,rect(t,0),'r','linewidth',2.5);
100 axis([-5 5 -3 3]);
101 legend('unit rect function');
102 xlabel('t');
103 ylabel('rect(t)');
104 grid minor on;
105
106
107 subplot(2,1,2)
108 plot(t,rect(t,0.5),'r','linewidth',2.5);
109 axis([-5 5 -3 3]);
110 legend('Unit Rect advanced by 0.5 units');
111 xlabel('t');
112 ylabel('rect(t+0.5)');
```

```
113 grid minor on;
114 saveas(gcf, 'plots/plot06.pdf')
115
116 figure(7)
117 subplot(2,1,1)
118 plot(t,rect(t,0),'r', 'linewidth',2.5);
119 axis([-5 5 -3 3]);
120 legend('unit rect function');
121 xlabel('t');
122 ylabel('rect(t)');
123 grid minor on;
124
125 subplot(2,1,2)
126 plot(t,rect(t,-0.5),'r', 'linewidth',2.5);
127 axis([-5 5 -3 3]);
128 legend('Unit rect delayed by 0.5 units');
129 xlabel('t');
130 ylabel('rect(t-0.5)');
131 grid minor on;
132 saveas(gcf, 'plots/plot07.pdf')
133
134
135 figure(8)
136 subplot(2,1,1)
137 plot(t,rect(t,0),'r', 'linewidth',2.5);
138 axis([-5 5 -3 3]);
139 legend('unit rect function');
140 xlabel('t');
141 ylabel('rect(t)');
142 grid minor on;
143
144 subplot(2,1,2)
145 plot(t,rect(t/3,0),'r', 'linewidth',2.5);
146 axis([-5 5 -3 3]);
147 legend('unit rect function with time scalling (expand)');
148 xlabel('t');
149 ylabel('rect(t/3)');
150 grid minor on;
151 saveas(gcf, 'plots/plot08.pdf')
152
153 figure(9)
154 subplot(2,1,1)
155 plot(t,rect(t,0),'r', 'linewidth',2.5);
156 axis([-5 5 -3 3]);
157 legend('unit rect function');
```

```
158 xlabel('t');
159 ylabel('rect(t)');
160 grid minor on;
161
162 subplot(2,1,2)
163 plot(t,rect(2*t,0),'r', 'linewidth',2.5);
164 axis([-5 5 -3 3]);
165 legend('unit rect function with time scalling (shrink)');
166 xlabel('t');
167 ylabel('rect(2t)');
168 grid minor on;
169 saveas(gcf,'plots/plot09.pdf')
170
171 figure(10)
172
173 subplot(2,1,1)
174 plot(t,ustep(t,0),'r', 'linewidth',2.5);
175 axis([-5 5 -3 5]);
176 legend('Unit Step');
177 xlabel('t');
178 ylabel('u(t)');
179 grid minor on;
180
181 subplot(2,1,2)
182 plot(t,ustep(-t,0),'r', 'linewidth',2.5);
183 axis([-5 5 -3 5]);
184 legend('Unit Step with time reversal');
185 xlabel('t');
186 ylabel('u(-t)');
187 grid minor on;
188 saveas(gcf,'plots/plot10.pdf')
189
190 % Plot of any arbiterary shap
191 y1 = ramp(t, 2, 2.5);
192 y2 = ramp(t, -5, 0);
193 y3 = ramp(t, 3, -2);
194 y4 = ustep(t,-4);
195
196 y = y1 + y2 + y3 + y4;
197
198 figure(11)
199
200 plot(t,y,'r', 'linewidth',2.5);
201 axis([-5 5 -3 5]);
202 xlabel('t');
```

```
203 ylabel('y(t)');  
204 grid minor on;  
205  
206 saveas(gcf, 'plots/plot11.pdf')  
207  
208 %saveas(gcf, 'plots/filename.pdf')
```

3.5 MATLAB Generated plots for $u(t)$, $rect(t)$ and their time-shift, time-scale and time-reversal

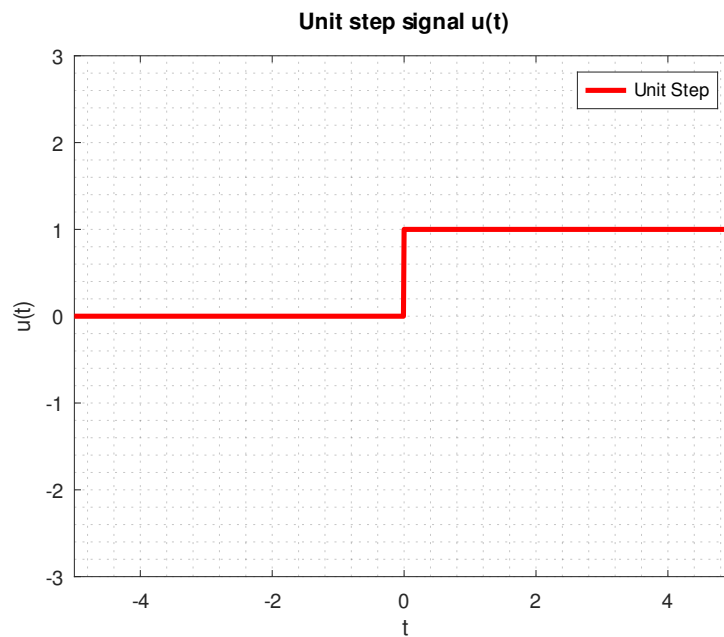


Figure 7: Unit step function

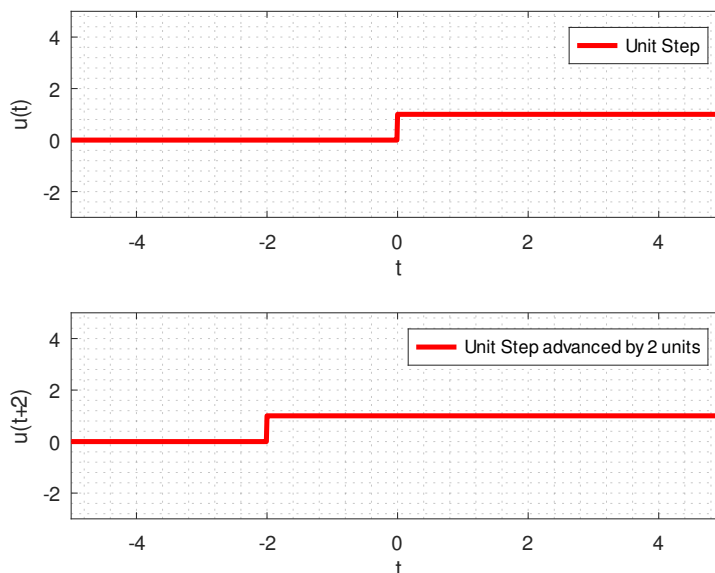


Figure 8: Unit step function with time shift (advanced)

4 Lab 04: Lab Tasks

1. Consider a finite-support signal

$$w(t) = \begin{cases} t, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Use MATLAB to plot

- (a) $w(t)$.
 - (b) $w(t+1)$.
 - (c) $w(-t+1)$.
 - (d) Add the above two signals in (b) and (c) to get a new signal $\gamma(t)$.
2. Consider the following signal $x(t)$ in Figure 18. Here we would like to consider expanded and compressed versions of that signal.
 - (a) Plot $x(t)$
 - (b) Plot $x(2t)$ and determine if it is a compressed or expanded version of $x(t)$.
 - (c) Plot $x(t/2)$ and determine if it is a compressed or expanded version of $x(t)$.
 - (d) $w(-t+1)$.
 3. Consider the analog signal $x(t)$ in Figure 19.

Plot the even-odd decomposition of $x(t)$ (i.e., find and plot the even $x_e(t)$ and the odd $x_o(t)$ components of $x(t)$).

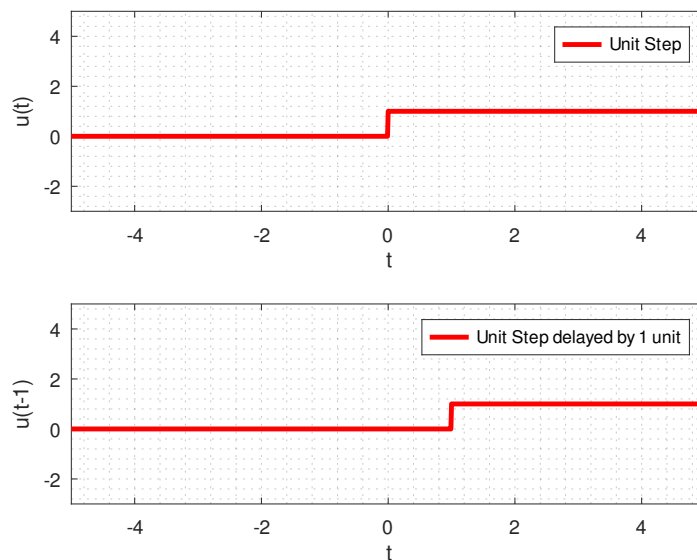


Figure 9: Unit step function with time shift (delayed)

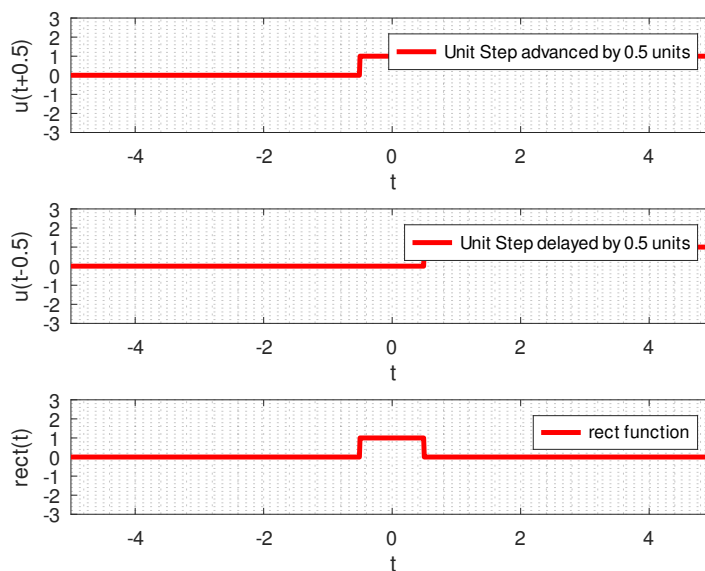


Figure 10: Generation of rect from two time-shifted unit-step functions

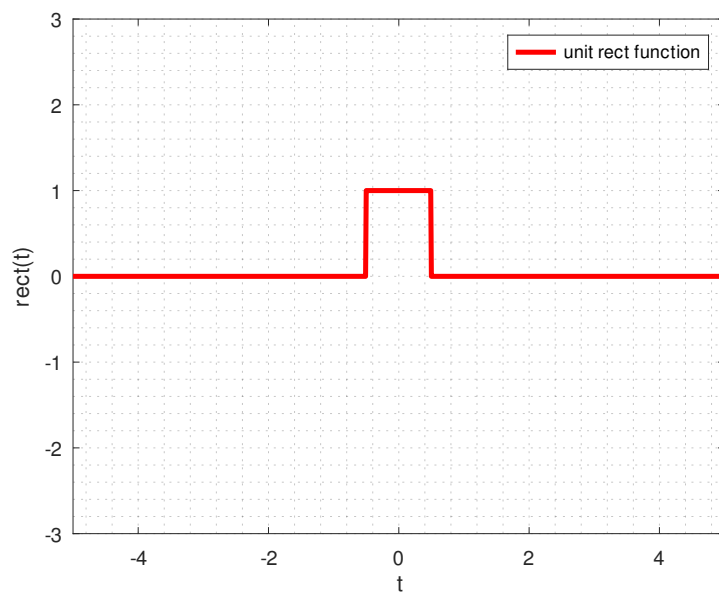


Figure 11: Generation of rect

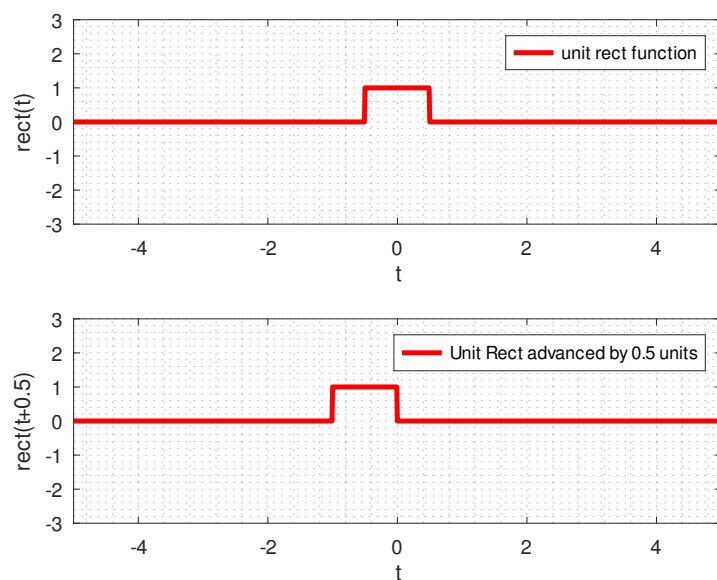


Figure 12: Generation of rect and its time-shifted (advanced) version

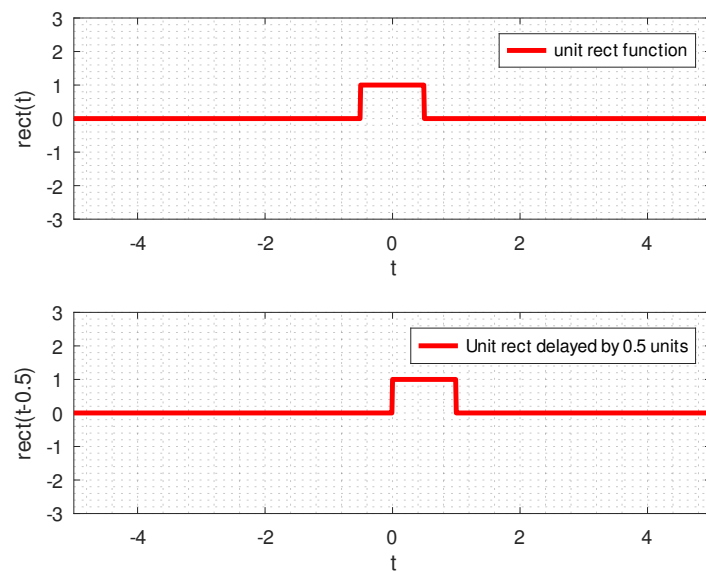


Figure 13: Generation of rect and its time-shifted (delayed) version

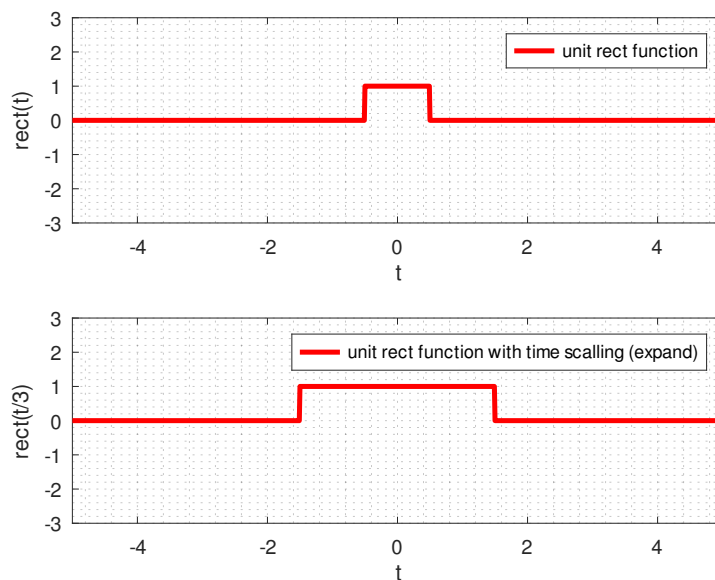


Figure 14: Generation of rect and its time-scaled (expanded) version

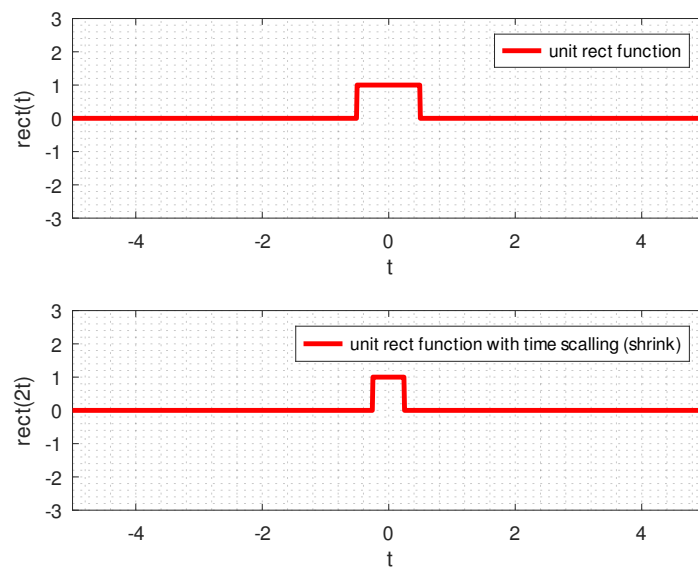


Figure 15: Generation of rect and its time-scaled (shrink) version

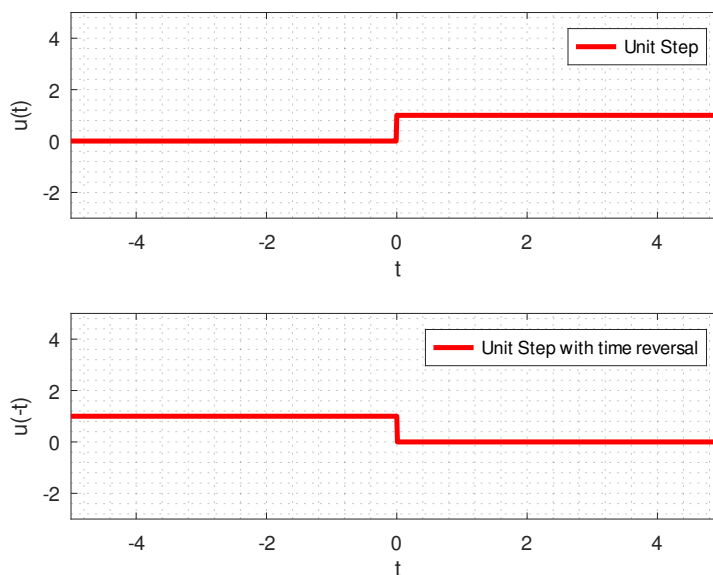


Figure 16: Generation of unit-step and its time-reverse version

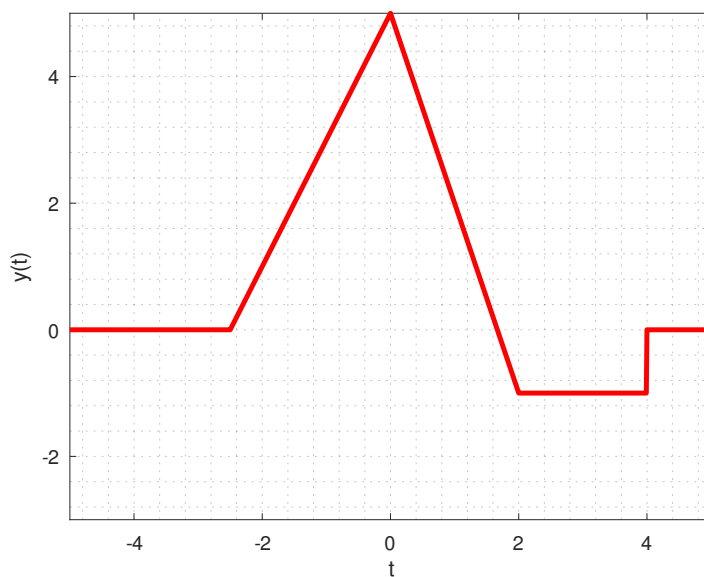
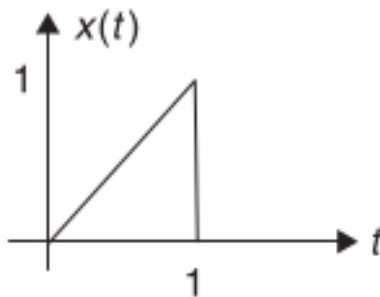
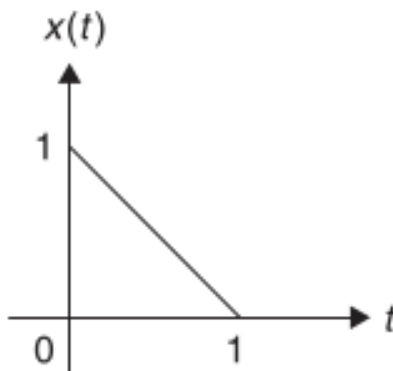


Figure 17: Generation of unit-step and its time-reverse version

Figure 18: $x(t)$ for task 2Figure 19: $x(t)$ for task 3